Discovering Computer Science introduces high school students to the fundamentals principles of computer science, emphasizing its role as a tool for problem-solving, communication, and personal expression. This course highlights the visible and impactful aspects of computing, encouraging students to explore how computer science influences the world around them. Students will engage with the design process, understand how data can solve widespread issues, and learn how physical computing with circuit boards can facilitate various input and output functions.

The course aims to provide students with a comprehensive understanding of computer science as a critical component of modern education. Through hands-on projects and real-world applications, students will develop the skills needed to become active contributors to our increasingly technological society. Whether they pursue careers in technology or not, this course equips students with the knowledge and tools to interpret and influence the digital world, fostering an appreciation for the profound impact of computer science on everyday life.

Additional Information

Computing is so fundamental to understanding and participating in society that it is valuable for every student to learn as part of a modern education. Computer science can be viewed as a liberal art, a subject that provides students with a critical lens for interpreting the world around them. Computer science prepares all students to be active and informed contributors to our increasingly technological society whether they pursue careers in technology or not. Computer science can be life-changing, not just skill training.

Students learn best when they are intrinsically motivated. This course prioritizes active learning experiences, relevant to students' lives, and provides students with authentic choice. Students are encouraged to be curious, solve personally relevant problems and express themselves through creation. Learning is an inherently social activity, so the course is designed to interweave lessons with discussions, presentations, peer feedback, and shared reflections. As students proceed through the pathway, the structures increasingly shift responsibility to students to formulate their own questions, develop their own solutions, and critique their work.

It is also critical to diversify the technology workforce. Addressing inequities within the field of computer science is critical to bringing computer science to all students. The tools and strategies in this course will help teachers understand and address well-known equity gaps within the field. All students can succeed in computer science when given the right support and opportunities, regardless of prior knowledge.

Additional Notes - Pedagogical Approach to Learning: Teacher as Lead Learner

What is the Lead Learner approach?

As the lead learner, the teacher's role shifts from being the source of knowledge to that of a leader in seeking knowledge. The lead learner's mantra is: "I may not know the answer, but I know that together we can figure it out."

The philosophy of the lead learner strategy is that students can benefit from having a model to demonstrate the learning process. Being a lead learner doesn't discount the need for a teacher to develop computer science content expertise, but it does allow for an environment of openness with students about the teacher learning process. Modeling and teaching *how to learn* are the most important factors to consider in order to be successful with this style of teaching and learning.

The lead learner technique represents good teaching practice in general. One important role of the teacher in the Computer Science Discoveries classroom is to model excitement about investigating how things work by asking motivating questions about why things work the way they do. With teacher guidance, students will learn how to hypothesize; ask questions of peers; test, evaluate, and refine solutions collaboratively; seek out resources; analyze data; and write clear and cogent code.

Discovering Computer Science (#0200384)

Version: 2025 - And Beyond (current)

Course Standards

Visit the specific benchmark webpage to find related instructional resources.

- SC.912.CC.1.1: Evaluate digital modes of communication and collaboration.

  Clarifications:
  Clarification 1: Instruction includes examples of appropriate tools including email, instant messaging, word processors and virtual meeting software.

- SC.912.CC.1.2: Utilize tools within a project environment to communicate.

  Clarifications:
  Clarification 1: Instruction includes discussing the productivity of each tool.

- SC.912.CC.1.3: Present information and data using presentation software.

  Clarifications:

  Clarification 1: Instruction includes computing devices such as probes, sensors, software tools, programs and handheld devices.

Clarification 2: Instruction includes analyzing and presenting interactive data visualizations.

- SC.912.CC.1.4: Create a digital artifact utilizing collaboration, reflection, analysis and iteration.
- SC.912.CC.2.1: Collaborate to publish information and data for a variety of audiences using digital tools and media-rich resources.
- SC.912.CC.2.2: Assess how collaboration influences the design and development of software artifacts.

  Clarifications:
  Clarification 1: Instruction includes comparing an individually designed project to a collaboratively designed project.

- SC.912.CC.2.3: Evaluate program designs and implementations for readability and usability.

  Clarifications:
  Clarification 1: Instruction includes evaluating programs done by third parties, peers and marketable programs.

- SC.912.CC.2.4: Critique the strengths and weaknesses of the collaborative process when creating digital products.
- SC.912.CO.1.1: Describe the efficiency and effectiveness of digital tools or resources used for real-world tasks.
- SC.912.CO.1.2: Identify and select the file format based on trade-offs.

  Clarifications:
  Clarification 1: Trade-offs for the most appropriate file format include analyzing the size, quality and accessibility of the file.

- SC.912.CO.1.4: Describe the relationship between drivers, hardware and operating systems.

  Clarifications:
  Clarification 1: Instruction focuses on the driver acting as a communication bridge between hardware and the operating system.

- SC.912.CO.1.5: Describe the organization of a computer and its principal components.

  Clarifications:
  Clarification 1: Instruction includes the identification of components by name, function and the interaction between them.

- SC.912.CO.1.7: Describe the process of protecting computer hardware from exploitation.

Clarifications:

Clarification 1: Instruction includes disabled unused USB ports, windows lock screens and managed access to restricted areas.

Clarification 2: Instruction includes the role of hardware such as tablets, phones and other devices.

Clarification 3: Instruction includes the role of cyber security software.

- SC.912.CO.1.8: Describe how the Internet facilitates global communication.

  Clarifications:
  Clarification 1: Instruction includes the grouping of smaller networks to provide a world wide web that facilitates communication.

- SC.912.CO.1.9: Evaluate the accuracy, relevance, comprehensiveness and bias of electronic information resources.
- SC.912.CO.2.3: Discuss the central processing unit (CPU).

  Clarifications:

  Clarification 1: Instruction includes similarities and differences between CPUs.

  Clarification 2: Instruction includes multi-core processing, hyper-threading and socket type.

- SC.912.CO.2.4: Explore the role of a power supply unit (PSU) in relation to a computer system.

  Clarifications:

  Clarification 1: Instruction includes wattage, modulation (semi, non and fully modular) and connector type.

  Clarification 2: Instruction includes the correct wattage for devices to prevent damage.

- SC.912.CO.2.7: Evaluate various forms of input and output (IO).

  Clarifications:
  Clarification 1: Instruction includes data used for or produced by input and output.

- SC.912.CO.2.8: Evaluate the basic components of wired computer networks.

  Clarifications:

  Clarification 1: Within this benchmark, components include a network interface card (NIC), an ethernet cable and a network switch.

  Clarification 2: Instruction includes two or more devices to communicate on a network.

- SC.912.CO.2.9: Evaluate the basic components of wireless computer networks.

  Clarifications:

  Clarification 1: Within this benchmark, components include a Wi-Fi adapter and wireless access point.

  Clarification 2: Instruction includes two or more devices to communicate on a network.

- SC.912.CO.2.11: Investigate the issues that impact network functionality.
- SC.912.CO.2.16: Describe how devices are identified on a network.

  Clarifications:
  Clarification 1: Instruction includes differentiation between public and private Internet protocol (IP) addresses.

- SC.912.CO.3.2: Develop criteria for selecting software when solving a specific real-world problem.

  Clarifications:
  Clarification 1: Instruction includes evaluating cost, features, reliability and usability.

- SC.912.CO.3.3: Examine the difference between operating system (OS) software and application software.

  Clarifications:
  Clarification 1: Instruction includes the role that operating systems play in relation to application programs.

- SC.912.CS.1.1: Identify possible risks to maintaining data confidentiality.

Clarifications:
Clarification 1: Within this benchmark, risks include shoulder surfing, illicit access to devices and theft of sensitive items.

- SC.912.CS.1.2: Describe computer security vulnerabilities.

  Clarifications:
  Clarification 1: Instruction includes student understanding that a computer worm can replicate itself across the network without human interaction while a computer virus requires human interaction to replicate.

- SC.912.CS.1.3: Evaluate computer security vulnerabilities.

  Clarifications:

  Clarification 1: Instruction includes evaluating the effects of attacks on computer systems.

  Clarification 2: Instruction includes evaluating the social and economic impacts on people.

- SC.912.CS.2.1: Analyze security and privacy issues that relate to computer networks and network connected devices.
- SC.912.CS.2.2: Describe security and privacy issues that relate to computer networks including the permanency of data on the Internet, online identity and privacy.
- SC.912.CS.3.4: Trace the social engineering attack cycle.

  Clarifications:

  Clarification 1: Instruction includes the various ways of collecting information, relationship building and how that information is used for exploitation.

  Clarification 2: Instruction includes discussing and evaluating the social and economic impact of the cycle on computer systems and people.

- SC.912.ET.1.1: Describe the emerging features of mobile devices, smart devices and vehicles.
- SC.912.ET.1.2: Describe the physical and cognitive challenges faced by users when learning to use computer interfaces.
- SC.912.ET.1.4: Examine device-to-device interactions that exclude human input.

Clarifications:
Clarification 1: Instruction includes making the connection to machine-to-machine (M2M) interaction.

- SC.912.ET.1.5: Explore the concepts of virtual and augmented reality.
- SC.912.ET.1.7: Describe how technology has changed the way people build and manage organizations and how technology impacts personal life.
- SC.912.ET.2.1: Explore the history of Artificial Intelligence (AI).

  Clarifications:

  Clarification 1: Instruction includes the application of AI tests.

  Clarification 2: Instruction includes how these tests have evolved along with AI.

- SC.912.ET.2.5: Describe major applications of artificial intelligence (AI) and machine learning.

  Clarifications:
  Clarification 1: Instruction includes discussing the applications to the medical, space and automotive fields.

- SC.912.ET.2.6: Describe how predictive Artificial Intelligence (AI) can be used to solve problems.

  Clarifications:
  Clarification 1: Instruction includes using predictive Artificial Intelligence (AI) to forecast trends, such as sports, the stock market and weather.

- SC.912.ET.3.2: Examine how robotics are used to address human challenges.
- SC.912.HS.1.1: Identify potential dangers to an individual's safety and security online.

  Clarifications:

  Clarification 1: Instruction includes the use of email, chat rooms and other forms of direct electronic communication.

  Clarification 2: Instruction includes the dangers of direct electronic communication including predatory behavior and human trafficking.

- SC.912.HS.1.2: Evaluate the consequences of cyberbullying.

Clarifications:

Clarification 1: Instruction includes the consequences for an individual engaged in bullying behavior.

Clarification 2: Instruction includes the consequences suffered by the victim of cyberbullying.

Clarification 3: Instruction includes the Jeffrey Johnson Stand Up for All Students Act.

- SC.912.HS.1.3: Determine the consequences of inaction when witnessing unsafe Internet practices.

  Clarifications:
  Clarification 1: Instruction focuses on the possible outcomes when suspicious Internet activity is not reported.

- SC.912.HS.1.4: Examine the positive outcomes when someone reports suspicious behavior on the Internet.

  Clarifications:

  Clarification 1: Instruction focuses on positive outcomes when action is taken relating to Internet reporting.

  Clarification 2: Instruction includes reporting to parents, school staff and peers.

- SC.912.HS.1.5: Evaluate the risks to personal information while accessing the Internet.

  Clarifications:

  Clarification 1: Instruction includes access to software, websites or web applications that do not protect against the disclosure, use or dissemination of an individual's personal information.

  Clarification 2: Instruction includes theft of personal data including social security numbers, banking information and identity.

Examples:
Example: John was on a new questionable website the previous night playing video games. When he woke up the following morning, he discovered that his email address had 30 new spam emails advertising various products. John noticed that his checking account

also had $20 missing from unauthorized charges. What do you think happened and what should his following steps be?

- SC.912.HS.1.6: Describe the impact of permissible privacy and security.

  Clarifications:
  Clarification 1: Instruction includes, but is not limited to, discussing privacy and security as it relates to account settings, cookies and application permissions.

- SC.912.HS.1.7: Construct strategies to combat cyberbullying or online harassment.
- SC.912.HS.2.1: Prioritize regulating screen time and the use of electronic devices for mental and physical well-being.

  Clarifications:
  Clarification 1: Instruction includes the role of digital media and communication, gaming devices, cellular devices, television and other digital sources as they relate to mental and physical well-being.

- SC.912.HS.2.2: Investigate the correlation between sedentary behavior and digital device use.
- SC.912.HS.2.3: Assess the role of digital health trackers in promoting healthy behaviors.
- SC.912.HS.2.4: Analyze the relationship between eye strain related to use of technology and exposure to increased blue light.

  Clarifications:
  Clarification 1: Instruction focuses on blurred vision, headaches, sleep deprivation and eye fatigue.

- SC.912.HS.3.1: Discuss the permanency of data on the Internet.

  Clarifications:

  Clarification 1: Instruction includes the permanency of sharing materials through digital communication and how it can affect future jobs, scholarship opportunities and potential positions.

  Clarification 2: Instruction focuses on confirmation of legitimacy before interacting with information from others, including liking, sharing and reposting.

- SC.912.HS.3.2: Analyze how social media influences the digital footprint of individuals, communities and cultures.
- SC.912.PE.1.1: Write code segments.

Clarifications:
Clarification 1: Instruction includes writing code segments that accept arguments and other segments such as functions, subroutines and methods.

- SC.912.PE.1.2: Create iterative and non-iterative structures within a program.

  Clarifications:
  Clarification 1: Iterative structures include nested iterative structures.

- SC.912.PE.1.3: Create selection structures within a program.

  Clarifications:
  Clarification 1: Instruction includes explaining selection structures and their uses within a program.

- SC.912.PE.1.10: Write programs that validate user input.
- SC.912.PE.1.12: Classify programming languages.

  Clarifications:

  Clarification 1: Instruction includes the classification of paradigms by object-oriented and procedural.

  Clarification 2: Instruction includes the application of domains by scientific applications and commercial applications.

- SC.912.PE.1.13: Describe and identify types of programming errors.

  Clarifications:
  Clarification 1: Instruction includes syntax, logic, runtime and computation errors.

- SC.912.PE.1.15: Implement a program using an integrated development environment (IDE) commonly used.
- SC.912.PE.1.17: Examine the building blocks of algorithms.

  Clarifications:
  Clarification 1: Building blocks include sequence, selection, iteration and recursion.

- SC.912.PE.1.18: Develop a computer program.

  Clarifications:

  Clarification 1: Instruction includes meeting the requirements set by a plan.

Clarification 2: Instruction includes the use of the software development cycle.

- SC.912.PE.1.19: Review a computer program to verify program functionality, programming styles, program usability and adherence to common programming standards.

  Clarifications:

  Clarification 1: Instruction includes peer review.

  Clarification 2: Instruction includes adherence to a programming language style guide.

- SC.912.PE.1.20: Write programs that use standard logic operators.
- SC.912.PE.1.21: Use Boolean logic to perform logical operations.
- SC.912.PE.1.23: Compile, run, test and debug a digital artifact.
- SC.912.PE.2.3: Compare techniques for analyzing massive data collections.
- SC.912.PE.3.1: Evaluate arithmetic expressions using operator precedence.
- SC.912.PE.3.2: Decompose a problem by defining new code segments.
- SC.912.PE.3.4: Evaluate algorithms by their efficiency, correctness and clarity.

  Clarifications:

  Clarification 1: Instruction includes analyzing and comparing execution times, testing with multiple inputs or data sets and debugging.

  Clarification 2: Instruction includes evaluating a well-known algorithm and implementing a new one.

  Clarification 3: Instruction includes comparing the efficiency between two or more algorithms.

- SC.912.PE.3.11: Perform advanced searches to locate information and design a data-collection approach to gather original data.
- SC.912.PE.3.14: Analyze data by identifying patterns through modeling and simulation of real-world data.
- SC.912.PE.3.15: Test the accuracy of scientific hypotheses using computer models and simulations.

- SC.912.PE.3.16: Design a representation of a computer program.

  Clarifications:

  Clarification 1: Instruction includes creating a plan that defines requirements, structural design, time estimates and testing elements.

  Clarification 2: Instruction includes the use of the software development cycle.

- SC.912.PE.3.18: Explain the principles of cryptography.

  Clarifications:
  Clarification 1: Instruction includes the principles of confidentiality, integrity, authentication, non-repudiation and key management.

- SC.912.PE.4.5: Define user prompts for clarity and usability within a program.

  Examples:
  Example: Mrs. Jan has given her programming class the task of creating an age verification application. Paolo, a student in the class, initially runs into an issue where when a user enters their birth month out, the program crashes. Define a user prompt that will correct this issue.

- SC.912.PE.4.6: Write a program that utilizes both input and output.

  Clarifications:
  Clarification 1: Instruction includes the end user entering the input and the program delivering the output.

- SC.912.PE.4.7: Use internal documentation to collaboratively design a program according to accepted standards.

  Clarifications:
  Clarification 1: Instruction includes multiple creators communicating within a program utilizing “clean code.”

- SC.912.TI.1.1: Analyze historical trends in hardware and software.

  Clarifications:

  Clarification 1: Instruction includes discussing technology upgrades for power requirements, computation capacity, speed, size, Artificial Intelligence (AI) and ease of use.

  Clarification 2: Instruction includes assessing the implications of technology trends on future computing devices.

- SC.912.TI.1.2: Identify ways to use technology to support lifelong learning.

  Clarifications:
  Clarification 1: Instruction includes the use of online tutorials, Artificial Intelligence (AI) and web searches to facilitate personal learning.

- SC.912.TI.1.3: Analyze the impact of digital media.

  Clarifications:

  Clarification 1: Instruction includes the analysis of digital media for implicit or explicit bias

  Clarification 2: Instruction includes discerning fact from opinion within digital media.

- SC.912.TI.1.4: Analyze the impact of digital media on culture and persona.

  Clarifications:
  Clarification 1: Instruction includes the effects of digital media on self- image and societal changes.

- SC.912.TI.1.5: Describe the impact of computing on business and commerce.
- SC.912.TI.1.6: Describe how technology impacts personal life.

  Clarifications:
  Clarification 1: Instruction includes evaluating the impact of smartwatches and various Internet of Things (IoT) devices.

- SC.912.TI.1.7: Evaluate ways in which technology may improve accessibility for the varying needs of learners, including students with disabilities (SWD).

  Clarifications:
  Clarification 1: Instruction includes assistive and instructional technologies.

- SC.912.TI.1.8: Explain how economic and societal factors are affected by access to critical information.
- SC.912.TI.1.9: Evaluate access and distribution of technology in a global society.

  Clarifications:
  Clarification 1: Instruction includes providing possible solutions to the challenges to equal access and the distribution of technology.

- SC.912.TI.1.10: Analyze technology-related career paths.

  Clarifications:

  Clarification 1: Technological career paths include programming, medical, health information technology (IT) and various other upcoming industries.

  Clarification 2: Instruction includes predicting future technology-related career trends.

- SC.912.TI.1.12: Examine the history of networking devices.

  Clarifications:
  Clarification 1: Instruction includes hubs, switches, ethernet cabling, wireless technology and fiber optics.

- SC.912.TI.1.13: Examine the historical impact of social media.

  Clarifications:

  Clarification 1: Instruction includes discussing the purpose of social media.

  Clarification 2: Instruction includes analyzing the impact of current social media platforms.

- SC.912.TI.2.1: Research how social media and technology can be used to distort, exaggerate or misrepresent information.

  Clarifications:

  Clarification 1: Instruction includes the consequences associated with posting misinformation, such as slander, libel and defamation.

  Clarification 2: Within this benchmark, emphasis should be placed on the impact of misinformation (clickbait, gaslighting, fake news, propaganda and deepfakes) on individuals, communities and cultures.

- SC.912.TI.2.2: Demonstrate knowledge of the Internet safety policy as it applies to state and district guidelines.

  Clarifications:

  Clarification 1: Instruction focuses on the current school district guidelines in which the student is enrolled.

Clarification 2: Instruction includes local and state level statutory requirements that govern Internet use.

- SC.912.TI.2.3: Recognize the terms and policies associated with the use of public access points.

  Clarifications:

  Clarification 1: Instruction includes understanding that using public access points may pose security risks.

  Clarification 2: Instruction includes discussing the importance of reading the full terms and conditions when using public access points.

- SC.912.TI.2.4: Explore the legal ramifications of technology use.

  Clarifications:

  Clarification 1: Instruction includes differentiating between legal and ethical responsibility.

  Clarification 2: Instruction includes understanding the importance of staying current with legal changes.

- SC.912.TI.2.5: Describe and model the legal use of modern communication media and devices.

  Clarifications:

  Clarification 1: Instruction includes the responsible use of modern communication media and devices.

  Clarification 2: Instruction includes discussion of personal safety when utilizing technology.

- SC.912.TI.2.6: Evaluate the impacts of the irresponsible use of information on collaborative projects.

  Clarifications:
  Clarification 1: Instruction includes discussing plagiarism, artificial intelligence (AI) chat usage and falsification of data.

- SC.912.TI.2.7: Describe differences between open source, freeware and proprietary software licenses and how they apply to different types of software.
- SC.912.TI.2.8: Evaluate the consequences of misrepresenting digital work as your own.

Clarifications:
Clarification 1: Instruction includes plagiarism, infringement and digital theft.

- SC.912.TI.2.9: Analyze how different categories of software licenses can be used to share and protect intellectual property.

  Clarifications:
  Clarification 1: Types of software licenses include open source and proprietary licenses.

- SC.912.TI.2.10: Analyze how access to information may not include the right to distribute the information.

  Clarifications:
  Clarification 1: Instruction includes comparing licensing in relation to ownership and distribution.

- SC.912.TI.2.11: Utilize citation tools when using digital information.
- SC.K12.CTR.1.1: **Actively participate in effortful learning both individually and collaboratively.**
  Students who actively participate in effortful learning both individually and with others:
  - Build perseverance by modifying methods as needed while solving a challenging task.
  - Stay engaged and maintain a positive mindset when working to solve tasks.
  - Help and support each other when attempting a new method or approach.

  Clarifications:
  Teachers who encourage students to participate actively in effortful learning both individually and with others:
  - Cultivate a community of learners.
  - Foster perseverance in students by choosing challenging tasks.
  - Recognize students' effort when solving challenging problems.
  - Emphasize project-based learning.
  - Establish a culture in which students ask questions of the teacher and their peers, and errors as a learning opportunity.
  - Develop students' ability to justify methods and compare their responses to the responses of their peers.
- SC.K12.CTR.2.1: **Demonstrate understanding by decomposing a problem.**

Students who demonstrate understanding by decomposing a problem:

- Analyze the problems in a way that makes sense given the task.
- Ask questions that will help with solving the task.
- Break down complex problems into individual problems.
- Decompose a complex problem into manageable parts.

Clarifications:

Teachers who encourage students to demonstrate understanding by decomposing a problem:

- Develop students' ability to analyze and problem-solve.
- Help students break complex tasks into subtasks.
- Show students that the solution to individual parts allows them to solve complex problems more effectively.

- SC.K12.CTR.3.1:

  Complete tasks with digital fluency.
  Students who complete tasks with digital fluency:Select and use appropriate digital tools by their functions.

  - Demonstrate proper typing techniques and keyboarding skills.
  - Understand responsible technology use.
  - Use feedback to improve efficiency using digital tools.
  - Relate previously learned concepts to new concepts.
  - Solve problems by developing, testing and refining technological processes.

  Clarifications:

  Teachers who encourage students to complete tasks with digital fluency:

  - Provide students with opportunities to increase critical thinking skills.
  - Provide students with opportunities to use various technology hardware and software, so that technology is an integral part of the learning experience. Develop students' ability to construct relationships between their current understanding and more sophisticated ways of thinking.

- SC.K12.CTR.4.1:

  Express solutions as computational steps.
  Students who express solutions as computational steps:

  - Solve problems step by step rather than all at once.
  - Represent solutions to problems in multiple ways, based on context or purpose.
  - Use patterns and structures to understand and connect computational concepts.
  - Check computations when solving problems.

  Clarifications:

  Teachers who encourage students to express solutions as computational steps:

  - Provide opportunities for students to develop sequentially based understandings of problems.
  - Guide students to align tasks to a step-by-step solution.
  - Select sequence and present student work to advance and deepen understanding of correct and increasingly efficient methods.
  - Prompt students to continually ask, “Does this solution make sense? How do you know?”
  - Reinforce that students check their work as they progress within and after a task.
  - Strengthen students’ ability to verify solutions through justification.

- SC.K12.CTR.5.1:

  Create an algorithm to achieve a given goal.
  Students who create algorithms to achieve a given goal:

  - Create or use a well-defined series of steps to achieve a desired outcome.
  - Compare the efficiency of an algorithm to those expressed by others.
  - Design a sequence of steps to follow.
  - Verify possible solutions by explaining the program or methods used.

  Clarifications:

Teachers who encourage students to create an algorithm to achieve a given goal:

- Support students to develop generalizations based on the similarities found among problems.
- Have students estimate or predict solutions before solving.
- Help students recognize the patterns in the world around them and connect these patterns to other concepts.

- SC.K12.CTR.6.1:

  Differentiate between usable data and miscellaneous information.
  Students who differentiate between usable data and miscellaneous information:

  - Express connections between concepts and representations.
  - Construct possible arguments based on evidence.
  - Perform decision-making between two actions.
  - Practice evaluating information and sources.
  - Perform investigations to gather data or determine if a program or method is appropriate.
  - Discern relevant, meaningful data from irrelevant or extraneous information.
  - Understand the characteristics and criteria determining whether data is relevant to a specific problem or task.

  Clarifications:

  Teachers who encourage students to differentiate between useable data and miscellaneous information:

  - Support students as they validate conclusions by comparing them to the given situation.
  - Create opportunities for students to discuss their thinking with peers.

- SC.K12.CTR.7.1:

  Solve real-life problems in science and engineering using computational thinking.
  Students who solve real-life problems in science and engineering using computational thinking:

- Adapt procedures to find solutions and apply them to a new context.
- Look for similarities among problems.
- Connect solutions of problems to more complicated large-scale situations.
- Connect concepts to everyday experiences.
- Use programs, models and methods to understand, represent and solve problems.
- Indicate how various concepts can be applied to other disciplines.
- Redesign programs, models and methods to improve accuracy or efficiency. Evaluate results based on the given context.

Clarifications:

Teachers who encourage students to solve real-life problems in science and engineering using computational thinking:

- Create learning opportunities that require logical reasoning and problem-solving skills.
- Provide opportunities for students to create plans and procedures to solve problems.
- Provide opportunities for students to create programs or models, both concrete and abstract, and perform investigations.
- Challenge students to question the accuracy of their programs, models and methods.

- MA.K12.MTR.1.1: **Actively participate in effortful learning both individually and collectively.**

  Mathematicians who participate in effortful learning both individually and with others:

  - Analyze the problem in a way that makes sense given the task.
  - Ask questions that will help with solving the task.
  - Build perseverance by modifying methods as needed while solving a challenging task.
  - Stay engaged and maintain a positive mindset when working to solve tasks.
  - Help and support each other when attempting a new method or approach.

Clarifications:
Teachers who encourage students to participate actively in effortful learning both individually and with others:

- Cultivate a community of growth mindset learners.
- Foster perseverance in students by choosing tasks that are challenging.
- Develop students' ability to analyze and problem solve.
- Recognize students' effort when solving challenging problems.

- MA.K12.MTR.2.1: **Demonstrate understanding by representing problems in multiple ways.**

  Mathematicians who demonstrate understanding by representing problems in multiple ways:

  - Build understanding through modeling and using manipulatives.
  - Represent solutions to problems in multiple ways using objects, drawings, tables, graphs and equations.
  - Progress from modeling problems with objects and drawings to using algorithms and equations.
  - Express connections between concepts and representations.
  - Choose a representation based on the given context or purpose.

  Clarifications:
  Teachers who encourage students to demonstrate understanding by representing problems in multiple ways:

  - Help students make connections between concepts and representations.
  - Provide opportunities for students to use manipulatives when investigating concepts.
  - Guide students from concrete to pictorial to abstract representations as understanding progresses.
  - Show students that various representations can have different purposes and can be useful in different situations.

- MA.K12.MTR.3.1: **Complete tasks with mathematical fluency.**

Mathematicians who complete tasks with mathematical fluency:

- Select efficient and appropriate methods for solving problems within the given context.
- Maintain flexibility and accuracy while performing procedures and mental calculations.
- Complete tasks accurately and with confidence.
- Adapt procedures to apply them to a new context.
- Use feedback to improve efficiency when performing calculations.

Clarifications:
Teachers who encourage students to complete tasks with mathematical fluency:

- Provide students with the flexibility to solve problems by selecting a procedure that allows them to solve efficiently and accurately.
- Offer multiple opportunities for students to practice efficient and generalizable methods.
- Provide opportunities for students to reflect on the method they used and determine if a more efficient method could have been used.

- MA.K12.MTR.4.1: **Engage in discussions that reflect on the mathematical thinking of self and others.**

  Mathematicians who engage in discussions that reflect on the mathematical thinking of self and others:

  - Communicate mathematical ideas, vocabulary and methods effectively.
  - Analyze the mathematical thinking of others.
  - Compare the efficiency of a method to those expressed by others.
  - Recognize errors and suggest how to correctly solve the task.
  - Justify results by explaining methods and processes.
  - Construct possible arguments based on evidence.

  Clarifications:
  Teachers who encourage students to engage in discussions that reflect on the mathematical thinking of self and others:

    - Establish a culture in which students ask questions of the teacher and their peers, and error is an opportunity for learning.
    - Create opportunities for students to discuss their thinking with peers.
    - Select, sequence and present student work to advance and deepen understanding of correct and increasingly efficient methods.
    - Develop students' ability to justify methods and compare their responses to the responses of their peers.
- MA.K12.MTR.5.1: **Use patterns and structure to help understand and connect mathematical concepts.**

  Mathematicians who use patterns and structure to help understand and connect mathematical concepts:

    - Focus on relevant details within a problem.
    - Create plans and procedures to logically order events, steps or ideas to solve problems.
    - Decompose a complex problem into manageable parts.
    - Relate previously learned concepts to new concepts.
    - Look for similarities among problems.
    - Connect solutions of problems to more complicated large-scale situations.

  Clarifications:
  Teachers who encourage students to use patterns and structure to help understand and connect mathematical concepts:

    - Help students recognize the patterns in the world around them and connect these patterns to mathematical concepts.
    - Support students to develop generalizations based on the similarities found among problems.
    - Provide opportunities for students to create plans and procedures to solve problems.
    - Develop students' ability to construct relationships between their current understanding and more sophisticated ways of thinking.
- MA.K12.MTR.6.1: **Assess the reasonableness of solutions.**

  Mathematicians who assess the reasonableness of solutions:

    - Estimate to discover possible solutions.

    - Use benchmark quantities to determine if a solution makes sense.
    - Check calculations when solving problems.
    - Verify possible solutions by explaining the methods used.
    - Evaluate results based on the given context.

  Clarifications:
  Teachers who encourage students to assess the reasonableness of solutions:

    - Have students estimate or predict solutions prior to solving.
    - Prompt students to continually ask, "Does this solution make sense? How do you know?"
    - Reinforce that students check their work as they progress within and after a task.
    - Strengthen students' ability to verify solutions through justifications.

- MA.K12.MTR.7.1: **Apply mathematics to real-world contexts.**

  Mathematicians who apply mathematics to real-world contexts:

    - Connect mathematical concepts to everyday experiences.
    - Use models and methods to understand, represent and solve problems.
    - Perform investigations to gather data or determine if a method is appropriate.
    - Redesign models and methods to improve accuracy or efficiency.

  Clarifications:
  Teachers who encourage students to apply mathematics to real-world contexts:

    - Provide opportunities for students to create models, both concrete and abstract, and perform investigations.
    - Challenge students to question the accuracy of their models and methods.
    - Support students as they validate conclusions by comparing them to the given situation.
    - Indicate how various concepts can be applied to other disciplines.

- ELA.K12.EE.1.1: Cite evidence to explain and justify reasoning.

Clarifications:
K-1 Students include textual evidence in their oral communication with guidance and support from adults. The evidence can consist of details from the text without naming the text. During 1st grade, students learn how to incorporate the evidence in their writing.

2-3 Students include relevant textual evidence in their written and oral communication. Students should name the text when they refer to it. In 3rd grade, students should use a combination of direct and indirect citations.

4-5 Students continue with previous skills and reference comments made by speakers and peers. Students cite texts that they've directly quoted, paraphrased, or used for information. When writing, students will use the form of citation dictated by the instructor or the style guide referenced by the instructor.

6-8 Students continue with previous skills and use a style guide to create a proper citation.

9-12 Students continue with previous skills and should be aware of existing style guides and the ways in which they differ.

- ELA.K12.EE.2.1: Read and comprehend grade-level complex texts proficiently.

  Clarifications:
  See Text Complexity for grade-level complexity bands and a text complexity rubric.

- ELA.K12.EE.3.1: Make inferences to support comprehension.

  Clarifications:
  Students will make inferences before the words infer or inference are introduced. Kindergarten students will answer questions like "Why is the girl smiling?" or make predictions about what will happen based on the title page. Students will use the terms and apply them in 2nd grade and beyond.

- ELA.K12.EE.4.1: Use appropriate collaborative techniques and active listening skills when engaging in discussions in a variety of situations.

  Clarifications:
  In kindergarten, students learn to listen to one another respectfully.

  In grades 1-2, students build upon these skills by justifying what they are thinking. For example: "I think ________ because

_______." The collaborative conversations are becoming academic conversations.

In grades 3-12, students engage in academic conversations discussing claims and justifying their reasoning, refining and applying skills. Students build on ideas, propel the conversation, and support claims and counterclaims with evidence.

- ELA.K12.EE.5.1: Use the accepted rules governing a specific format to create quality work.

  Clarifications:
  Students will incorporate skills learned into work products to produce quality work. For students to incorporate these skills appropriately, they must receive instruction. A 3rd grade student creating a poster board display must have instruction in how to effectively present information to do quality work.

- ELA.K12.EE.6.1: Use appropriate voice and tone when speaking or writing.

  Clarifications:
  In kindergarten and 1st grade, students learn the difference between formal and informal language. For example, the way we talk to our friends differs from the way we speak to adults. In 2nd grade and beyond, students practice appropriate social and academic language to discuss texts.

- ELD.K12.ELL.SI.1: English language learners communicate for social and instructional purposes within the school setting.

Course Information

General Notes

This course should be taught using Florida's State Academic Standards for Computer Science: Florida's B.E.S.T. ELA Expectations (EE), Mathematical Thinking and Reasoning Standards (MTRs) and Computational Thinking and Reasoning Standards (CTRs) for students. Florida educators should intentionally embed these standards within the content and their instruction as applicable.

English Language Development ELD Standards Special Notes Section

Teachers are required to provide listening, speaking, reading and writing instruction that allows English language learners (ELL) to communicate information, ideas and concepts for academic success in the content area of Mathematics. For the given level of English language proficiency and with visual, graphic, or interactive support, students will interact with grade-level words, expressions, sentences and discourse to process or produce language necessary for academic success. The ELD standard should specify a relevant content area concept or topic of study chosen by curriculum developers and teachers which maximizes an ELL's need for communication and social skills. To access an ELL supporting document which delineates performance definitions and descriptors, please click on the following link: https://cpalmsmediaprod.blob.core.windows.net/uploads/docs/standards/eld/si.pdf.

Accommodations

Federal and state legislation requires the provision of accommodations for students with disabilities as identified on the secondary student's Individual Educational Plan (IEP) or 504 plan or postsecondary student's accommodations' plan to meet individual needs and ensure equal access. Accommodations change the way the student is instructed. Students with disabilities may need accommodations in such areas as instructional methods and materials, assignments and assessments, time demands and schedules, learning environment, assistive technology and special communication systems. Documentation of the accommodations requested and provided should be maintained in a confidential file.

In addition to accommodations, some secondary students with disabilities (students with an IEP served in Exceptional Student Education (ESE) will need modifications to meet their needs. Modifications change the outcomes and or what the student is expected to learn, e.g., modifying the curriculum of a secondary career and technical education course.

General Information

Course Number: 0200384
Course Path: **Section:** Grades PreK to 12 Education Courses > **Grade Group:** Grades 9 to 12 and Adult Education Courses > **Subject:** Computer Science > **SubSubject:** General >
Abbreviated Title: DISCOVERING CS
Course Length: Year (Y)
Course Level: 2
Course Status: Course Approved

Graduation Requirement: Electives

Educator Certifications

One of these educator certification options is required to teach this course.

- Computer Science (Elementary and Secondary Grades K-12)
- Classical Education - Restricted (Elementary and Secondary Grades K-12)

  Section 1012.55(5), F.S., authorizes the issuance of a classical education teaching certificate, upon the request of a classical school, to any applicant who fulfills the requirements of s. 1012.56(2)(a)-(f) and (11), F.S., and Rule 6A-4.004, F.A.C. Classical schools must meet the requirements outlined in s. 1012.55(5), F.S., and be listed in the FLDOE Master School ID database, to request a restricted classical education teaching certificate on behalf of an applicant.

Qualifications

As well as any certification requirements listed on the course description, the following qualifications may also be acceptable for the course:

Any field when certification reflects a bachelor or higher degree.